Problem 4.2

Q's: Do elec. potentials equal each other over salt bridge? (Flow of ions is permitted?)

(a) $\overset{\alpha}{Cu(s)} \mid \overset{\beta}{0.001\,M\ CuSO_4(aq)} \parallel \overset{\beta'}{0.1\,M\ CuSO_4(aq)} \mid \overset{\alpha'}{Cu(s)}$

$U_{cell} = V^{\alpha'} - V^{\alpha}$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium

LHS: $Cu \rightleftharpoons Cu^{2+} + 2e^- \Rightarrow \mu_{Cu}^{\alpha} = \mu_{Cu^{2+}}^{\beta} + 2\mu_e^{\alpha} \Rightarrow \mu_e^{\alpha} = \frac{1}{2}\left(\mu_{Cu}^{\alpha} - \mu_{Cu^{2+}}^{\beta}\right)$

RHS: $Cu \rightleftharpoons Cu^{2+} + 2e^- \Rightarrow \mu_{Cu}^{\alpha'} = \mu_{Cu^{2+}}^{\beta'} + 2\mu_e^{\alpha'} \Rightarrow \mu_e^{\alpha'} = \frac{1}{2}\left(\mu_{Cu}^{\alpha'} - \mu_{Cu^{2+}}^{\beta'}\right)$

$\therefore \boxed{U_{cell} = \frac{1}{2F}\left(\mu_{Cu}^{\alpha} - \mu_{Cu^{2+}}^{\beta} - \mu_{Cu}^{\alpha'} + \mu_{Cu^{2+}}^{\beta'}\right)}$

(b) $\overset{\alpha}{Ag(s)} \mid \overset{\beta}{AgCl(s)} \mid \overset{\beta'}{HCl(aq)} \mid \overset{\alpha'}{H_2(g)}, Pt(s)$

$U_{cell} = V^{\alpha'} - V^{\alpha}$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium : $\mu_e^{\alpha} = \mu_e^{\beta}$

LHS: $AgCl + e^- \rightleftharpoons Ag + Cl^- \Rightarrow \mu_{AgCl}^{\beta} + \mu_e^{\alpha} = \mu_{Ag}^{\alpha} + \mu_{Cl^-}^{\beta'} \Rightarrow \mu_e^{\alpha} = \mu_{Ag}^{\alpha} - \mu_{Cl^-}^{\beta'} - \mu_{AgCl}^{\beta}$

RHS: $H_2 \rightleftharpoons 2H^+ + 2e^- \Rightarrow \mu_{H_2}^{\alpha'} = 2\mu_{H^+}^{\beta'} + 2\mu_e^{\alpha'} \Rightarrow \mu_e^{\alpha'} = \frac{1}{2}\left(\mu_{H_2}^{\alpha} - 2\mu_{H^+}^{\beta'}\right)$

$\therefore \boxed{U_{cell} = \frac{1}{F}\left(\mu_{Ag}^{\alpha} - \mu_{Cl^-}^{\beta'} - \mu_{AgCl}^{\beta} - \frac{1}{2}\left(\mu_{H_2}^{\alpha} - 2\mu_{H^+}^{\beta'}\right)\right)}$

(c) $\overset{\alpha}{Zn(s)} \mid \overset{\beta}{ZnO(s)} \mid \overset{\beta'}{KOH(aq)} \mid \overset{\alpha'}{O_2(g)}, Pt(s)$

$U_{cell} = V^{\alpha'} - V^{\alpha}$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium : $\mu_e^{\alpha} = \mu_e^{\beta}$

LHS: $Zn + 2OH^- \rightleftharpoons ZnO + H_2O + 2e^- \Rightarrow \mu_{Zn}^{\alpha} + 2\mu_{OH^-}^{\beta'} = \mu_{ZnO}^{\beta} + \mu_{H_2O}^{\beta'} + 2\mu_e^{\alpha} \Rightarrow \mu_e^{\alpha} = \frac{1}{2}\left(\mu_{Zn}^{\alpha} + 2\mu_{OH^-}^{\beta'} - \mu_{ZnO}^{\beta} - \mu_{H_2O}^{\beta'}\right)$

RHS: $4OH^- \rightleftharpoons O_2 + 4e^- + 2H_2O \Rightarrow 4\mu_{OH^-}^{\beta'} = \mu_{O_2}^{\alpha'} + 4\mu_e^{\alpha'} + 2\mu_{H_2O}^{\beta'} \Rightarrow \mu_e^{\alpha'} = \frac{1}{4}\left(4\mu_{OH^-}^{\beta'} - \mu_{O_2}^{\alpha'} - 2\mu_{H_2O}^{\beta'}\right)$

$\therefore U_{cell} = \frac{1}{2F}\left(\mu_{Zn}^{\alpha} + 2\mu_{OH^-}^{\beta'} - \mu_{ZnO}^{\beta} - \mu_{H_2O}^{\beta'} - \frac{1}{2}\left(4\mu_{OH^-}^{\beta'} - \mu_{O_2}^{\alpha'} - 2\mu_{H_2O}^{\beta'}\right)\right) = \boxed{\frac{1}{2F}\left(\mu_{Zn}^{\alpha} - \mu_{ZnO}^{\beta} + \frac{1}{2}\mu_{O_2}^{\alpha'}\right)}$

(d) $\overset{\alpha}{Pt(s), O_2(g)} \mid \overset{\beta}{NaOH(aq)} \parallel NH_3(aq), \overset{\beta'}{NaNO_2(aq)} \mid \overset{\alpha'}{Pt(s)}$

$\mu_{H_2O} = ?$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium : $\mu_{H_2O}^{\beta} = \mu_{H_2O}^{\beta'}$;

LHS: $4OH^- \rightleftharpoons O_2 + 2H_2O + 4e^- \Rightarrow 4\mu_{OH^-}^{\beta} = \mu_{O_2}^{\alpha} + 2\mu_{H_2O}^{\beta} + 4\mu_e^{\alpha} \Rightarrow \mu_e^{\alpha} = \frac{1}{4}\left(4\mu_{OH^-}^{\beta} - \mu_{O_2}^{\alpha} - 2\mu_{H_2O}^{\beta}\right)$

RHS: $NO_2^- + 6e^- + 5H_2O \rightleftharpoons NH_3 + 7OH^- \Rightarrow \mu_{NO_2^-}^{\beta'} + 6\mu_e^{\alpha'} + 5\mu_{H_2O}^{\beta'} = \mu_{NH_3}^{\beta'} + 7\mu_{OH^-}^{\beta} \Rightarrow \mu_e^{\alpha'} = \frac{1}{6}\left(\mu_{NH_3}^{\beta'} + 7\mu_{OH^-}^{\beta} - 5\mu_{H_2O}^{\beta'} - \mu_{NO_2^-}^{\beta'}\right)$

$\therefore \boxed{U_{cell} = \frac{1}{2F}\left(\frac{1}{2}\left(4\mu_{OH^-}^{\beta} - \mu_{O_2}^{\alpha} - 2\mu_{H_2O}^{\beta}\right) - \frac{1}{3}\left(\mu_{NH_3}^{\beta'} + 7\mu_{OH^-}^{\beta} - 5\mu_{H_2O}^{\beta'} - \mu_{NO_2^-}^{\beta'}\right)\right)}$

Problem 4.3 $\overset{\alpha}{\quad} \overset{\beta}{\quad} \overset{\beta'}{\quad} \overset{\alpha'}{\quad}$

Problem 4.3

α β β' α'

$Pt_{(s)} \mid FeCl_{2(aq)}, FeCl_{3(aq)} \parallel SnCl_{2(aq)}, SnCl_{4(aq)} \mid Pt_{(s)}$

(a) + Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$

◦ Phases in adjacent phases are at equilibrium

LHS: $Fe^{3+} \rightleftharpoons Fe^{2+} + e^- \Rightarrow \mu_{Fe^{3+}}^{\beta} = \mu_{Fe^{2+}}^{\beta} + \mu_e^{\alpha} \Rightarrow \mu_e^{\alpha} = \mu_{Fe^{3+}}^{\beta} - \mu_{Fe^{2+}}^{\beta}$

RHS: $Sn^{2+} \rightleftharpoons Sn^{4+} + 2e^- \Rightarrow \mu_{Sn^{2+}}^{\beta'} = \mu_{Sn^{4+}}^{\beta'} + 2\mu_e^{\alpha'} \Rightarrow \mu_e^{\alpha'} = \frac{1}{2}\left(\mu_{Sn^{2+}}^{\beta'} - \mu_{Sn^{4+}}^{\beta'}\right)$

$\therefore U_{cell} = \frac{1}{F}\left(\mu_{Fe^{3+}}^{\beta} - \mu_{Fe^{2+}}^{\beta} - \frac{1}{2}\left(\mu_{Sn^{2+}}^{\beta'} - \mu_{Sn^{4+}}^{\beta'}\right)\right)$

(b) $\mu_{Fe^{2+}}^{\beta} = RT \ln\left(\lambda_{Fe^{2+}}^{\beta}\right) = RT \ln\left(C_{Fe^{2+}}^{\beta} f_{Fe^{2+}} a^{\circ}_{Fe^{2+}}\right)$

$\mu_{Fe^{3+}}^{\beta} = RT \ln\left(\lambda_{Fe^{3+}}^{\beta}\right) = RT \ln\left(C_{Fe^{3+}}^{\beta} f_{Fe^{3+}} a^{\circ}_{Fe^{3+}}\right)$

$\mu_{Sn^{2+}}^{\beta'} = RT \ln\left(\lambda_{Sn^{2+}}^{\beta'}\right) = RT \ln\left(C_{Sn^{2+}}^{\beta'} f_{Sn^{2+}} a^{\circ}_{Sn^{2+}}\right)$

$\mu_{Sn^{4+}}^{\beta'} = RT \ln\left(\lambda_{Sn^{4+}}^{\beta'}\right) = RT \ln\left(C_{Sn^{4+}}^{\beta'} f_{Sn^{4+}} a^{\circ}_{Sn^{4+}}\right)$

(c) $U_{cell} = \frac{1}{F}\left[\left(RT \ln\left(\frac{a_{Fe^{2+}}^{\beta}}{a_{Fe^{3+}}^{\beta}} \cdot \left(\frac{a_{Sn^{4+}}^{\beta'}}{a_{Sn^{2+}}^{\beta'}}\right)^{1/2}\right)\right) + RT \ln\left(\frac{C_{Fe^{2+}}^{\beta} f_{Fe^{2+}}^{\beta}}{C_{Fe^{3+}}^{\beta} f_{Fe^{3+}}^{\beta}} \cdot \frac{\left(C_{Sn^{4+}}^{\beta'} f_{Sn^{4+}}^{\beta'}\right)^{1/2}}{\left(C_{Sn^{2+}}^{\beta'} f_{Sn^{2+}}^{\beta'}\right)^{1/2}}\right)\right]$

Assuming $f_i = 1$:

$$\boxed{U_{cell} = \left[\left(\frac{RT}{F} \ln\left(\frac{a_{Fe^{2+}}^{\beta}}{a_{Fe^{3+}}^{\beta}} \cdot \left(\frac{a_{Sn^{4+}}^{\beta'}}{a_{Sn^{2+}}^{\beta'}}\right)^{1/2}\right)\right) + \frac{RT}{F} \ln\left(\frac{C_{Fe^{2+}}^{\beta}\left(C_{Sn^{4+}}^{\beta'}\right)^{1/2}}{C_{Fe^{3+}}^{\beta}\left(C_{Sn^{2+}}^{\beta'}\right)^{1/2}}\right)\right]}$$

U°_{cell} — Nernst correction

Ferric → 3+
Ferrous → 2+

(d) $(U_{cell} - U^{\circ}_{cell}) = \frac{RT}{F} \ln\left(\frac{1 \times 10^{-3}}{0.5 \times 10^{-3}} \left(\frac{0.2 \times 10^{-3}}{0.7 \times 10^{-3}}\right)^{1/2}\right)$

$\boxed{U_{cell} = 1.72 \text{ mV vs. standard cell potential}}$

Problem 4.7

Ni electrode ; V vs. pH diagram

(a) As shown in class:

Nernst correction for HER: $H^+ + e^- \rightleftharpoons \frac{1}{2}H_2$

$$\left(U_{H_2/H^+} - U^{\circ}_{H_2/H^+}\right) = \left(U^{\circ}_{H_2/H^+} - U^{\circ}_{H_2/H^+}\right) - \frac{RT}{F} \ln\left(\frac{P_H^{1/2}}{C_{H^+}}\right)$$

$$= -0.0591 (pH) \quad V \text{ vs. SHE}$$

Nernst correction for OER: $2H_2O \rightleftharpoons O_2 + 4H^+ + 4e^-$ ($2H_2O - O_2 - 4H^+ = 4e^-$)

$$\left(U_{O_2/H_2O} - U^{\circ}_{H_2/H^+}\right) = \left(U^{\circ}_{O_2/H_2O} - U^{\circ}_{H_2/H^+}\right) - \frac{RT}{4F} \ln\left(\frac{1}{P_{O_2} \cdot C_{H^+}^4}\right)$$

$$= 1.23 + \frac{RT}{F} \ln C_{H^+} \quad ; \quad pH = -\log H^+ \quad ; \quad \ln(a) = \frac{\log(a)}{\log(e)}$$

$$= 1.23 - 0.0591 (pH) \quad V \text{ vs. SHE} \qquad \ln(C_{H^+}) = (2.302)_{pH}$$

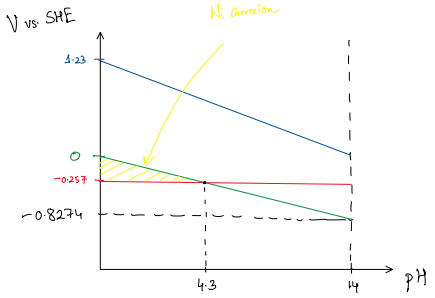
$Ni \rightleftharpoons Ni^{2+}_{(aq)} + 2e^-$

Nernst correction for Ni electrode : $\left(U_{Ni/Ni^{2+}} - U_{H_2/H^+}\right) = \left(U^{\circ}_{Ni/Ni^{2+}} - U_{H_2/H^+}\right) + \frac{RT}{2F} \ln\left(\frac{C_{Ni^{2+}}^{1}}{C_{Ni^{2+}}}\right)$

$Ni \rightleftharpoons Ni^{2+}_{(aq)} + 2e^-$

Nernst correction for Ni electrode : $(U_{Ni/Ni^{2+}} - U_{H_2/H^+}) = (U^\circ_{Ni/Ni^{2+}} - U_{H_2/H^+}) + \frac{RT}{2F} \ln\left(\frac{a_{Ni}}{C_{Ni^{2+}}}\right)$, with $a_{Ni} = 1$

$U_{Ni/Ni^{2+}} = -0.257 + \frac{RT}{F} \ln (C_{Ni^{2+}})^{1/2}$ ← Not dependent on pH



Stability: $(U_{Ni/Ni^{2+}} - U_{H_2/H^+}) < 0$, Non-spont @ pH < 4.3

↑ More negative (will be oxidized)

$(U_{Ni/Ni} - U_{H_2/H^+}) > 0$, Spontaneous , @ pH > 4.3

↑ More negative (will be oxidized)

∴ Ni will be oxidized / corrode below pH 4.3 in acidic solution

Problem 4.9

Cu electrode in $CuSO_4$ electrolyte ($Cu \rightleftharpoons Cu^{2+} + 2e^-$) ; 0.05M $[Cu^{2+}]$

(a) Calomel electrode ($Hg - Hg_2Cl_2$ RE)

$(U_{Cu/Cu^{2+}} - U_{Hg/HgCl}) = ?$

Nernst correction for Cu/Cu^{2+} ⇒ $U_{Cu/Cu^{2+}} - U_{H_2/H^+} = (U^\circ_{Cu/Cu^{2+}} - U_{H_2/H^+}) - \frac{RT}{2F} \ln C_{Cu^{2+}}^{-1}$

$= 0.337 + \frac{RT}{F} \ln (C_{Cu^{2+}})^{+1/2}$

$U_{Cu/Cu^{2+}} = 0.299$ vs. SHE @ 298 K and 0.05M $[Cu^{2+}]$

Since no $[Cl^-]$ variation : $U_{Hg/Hg_2Cl_2} = (U^\circ_{Hg/Hg_2Cl_2} - U^\circ_{H_2/H^+}) - \frac{0.0591}{2} \log C_{Cl^-}^2$ → The RE has its own electrolyte and therefore needs to corrected for depending on the electrolyte used

$= 0.244$ V vs. SHE @ 25°C and saturated solution [ChemLibreText]

∴ $(U_{Cu/Cu^{2+}} - U_{Hg/Hg_2Cl_2}) = 0.131$ V

$U_{Cu/Cu^{2+}} = 0.05$ V vs. calomel

(b) Doing a similar analysis :

$U_{Cu/Cu^{2+}} = 0.375$ vs. SHE

$U_{Ag/AgCl} = (U^\circ_{Ag/AgCl} - U_{H_2/H^+}) - 0.0591 \log C_{Cl^-}$ [ChemLibreText]

$U_{Ag/AgCl} = 0.197$ V vs. SHE @ 25°C and saturated solution

∴ $U_{Cu/Cu^{2+}} = 0.102$ V vs. Ag/AgCl RE

(c) Iron electrode. $Fe(CN)_6^{4-}_{(aq)} \rightleftharpoons Fe(CN)_6^{3-}_{(aq)} + e^-$

[10x] [x]

Nernst correction for iron electrode: $U_{Fe^{3+}/Fe^{2+}} - U_{H_2/H^+} = \left[U^\circ_{Fe^{3+}/Fe^{2+}} - U_{H_2/H^+}\right) - \frac{RT}{F} \ln\left(\frac{C_{Fe(CN)_6^{4-}}}{C_{Fe(CN)_6^{3-}}}\right)$

$= 0.358 - 0.0591 \log(10)$

$= 0.298$ V vs. SHE

∴ $U_{Fe^{3+}/Fe^{2+}} = 0.055$ V vs. Calomel

$U_{Fe^{3+}/Fe^{2+}} = 0.101$ V vs. Ag/AgCl RE

Problem 5.2

i for reduction of protons (H^+) → HER $H_2 - 2H^+ \rightleftharpoons 2e^-$; n=2,

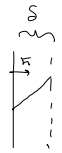
+ Rate controlled by MT of H^+ in diffusion layer ($\delta = 40\times10^{-6}$ m) → i.e. we are in the limiting current density regime

+ $D_{H^+} = 9.310 \times 10^{-9}$ m^2/s (@ www.agion.be)

+ $C_{H^+,\infty} = 0.01$ M

$$\bar{N}_{H^+} = -z_j u_j F c_j \nabla\phi - D_j \nabla c_j + c_j \bar{v}$$; 1D transport w/ diffusion (No migration) (No mvtj) (0)

$$\frac{i}{F} = \sum z_j \bar{N}_j$$ (0 (does not cont. to i))



$$\bar{n}\cdot\bar{N}_{H^+} = \frac{-s_j}{nF} i_n$$

$$i_n = \frac{nF}{s_j}\left(\bar{n}\cdot\bar{N}_{H^+}\right)$$

$$= \frac{2F}{(-2)}\left(-D_{H^+}\frac{\partial c}{\partial x}\right)$$

$$\frac{\partial c}{\partial x} = \frac{c_{bulk}}{\delta}$$ (Linear profile assumption)

$$i = +F\cdot D_{H^+}\cdot\frac{c_{bulk}}{\delta}$$

$$= (96486)\cdot(9.310\times10^{-9})\cdot\frac{(0.01/0.001)}{40\times10^{-6}}$$

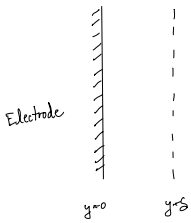
$\frac{C}{mol}\cdot\frac{z^+}{C}\cdot\frac{mol}{m^3}\cdot\frac{1}{m}$

$$i \approx 224.6\ A\,m^{-2}$$

Problem 8

$$PbO_{2(s)} + SO_{4(aq)}^{2-} + 4H^+_{(aq)} + 2e^- \rightleftharpoons PbSO_{4(s)} + 2H_2O$$

Binary electrolyte: H_2SO_4 → dissociating into $2H^+$ and $1\,SO_4^{2-}$



$z_+ = +1$ $z_- = -2$ $s_+ = -1$ $s_- = -1$

$\nu_+ = 2$ $\nu_- = 1$

$$c = \frac{c_+}{\nu_+} = \frac{c_-}{\nu_-}$$

Following class notes:

* Flux of each species:

1) $\nabla\cdot N_+ = \nabla\cdot\left(-z_+u_+Fc\nu_+ - D_+u_+\nabla c\right) + \nu_+\nabla c\cdot\bar{v}$

2) $\nabla\cdot N_- = \nabla\cdot\left(-z_-u_-Fc\nu_- - D_-u_-\nabla c\right) + \nu_-\nabla c\cdot\bar{v}$

* Material balance:

3) $\nu_+\frac{\partial c}{\partial t} = -\nabla\cdot N_+ \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_+}(\nabla\cdot N_+)$

4) $\nu_-\frac{\partial c}{\partial t} = -\nabla\cdot N_- \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_-}(\nabla\cdot N_-)$

* Subtracting ③ and ④ & rearranging: $\nabla \cdot (c \nabla \phi) = - \frac{(D_+ - D_-)}{F(z_+ u_+ - z_- u_-)} \nabla^2 c$

* Plugging into ③:

$$\frac{\partial c}{\partial t} = -\frac{1}{\nu_+} \left(\nabla \cdot \left[-z_j u_j F c_j \nabla \phi - D_j \nabla c_j + c_j \bar{v} \right] \right)$$

Divergence of vector = ?

$$= -\frac{1}{\nu_+} \left(-z_j u_j F c_j \nabla \cdot (\nabla \phi) - D_j \nabla \cdot (\nabla c_j) \right)$$

$$= z_j u_j \frac{c_j}{\nu_+} \nabla \cdot \nabla \phi - \frac{D_j}{\nu_+} \nabla \cdot (\nabla c_j)$$

$$= z_j u_j c \nabla \cdot \nabla \phi$$

$$\vdots$$

Nernst Einstein

$$\frac{\partial c}{\partial t} = D \nabla^2 c - \bar{v} \cdot \nabla c \quad ; \quad D = \frac{(z_+ u_+ D_- - z_- u_- D_+)}{z_+ D_+ - z_- D_-} = \frac{(z_+ - z_-) D_+ D_-}{z_+ D_+ - z_- D_-} = \frac{3 D_+ D_-}{D_+ + 2 D_-}$$

$i = \frac{nF}{-s_j} N_j$

↓

Solving this ODE for $\frac{\partial c}{\partial t} = 0$ (SS) & BC: 1) @ $y = 0$, $c = 0$ ∝ Both species are reacting at the electrode

2) @ $y = \delta$, $c = c_{bulk}$

$$c = \frac{c_{bulk}}{\delta} y$$

@ $x = 0$: $i = \frac{2F}{4} N_+$, $i = 2F N_-$ → $N_+ = N_- \cdot 4$

$N_- \neq 0$ → cannot solve for $\frac{d\phi}{dy}$

Need to use this relationship!

& $N_- = 0$ we can get $N_+ = -\nu_+ \frac{(z_- u_- D_+ - z_+ u_+ D_-)}{z_- u_-} \frac{dc}{dy}$

For H^+ : $z_+ = +1$, $\nu_+ = 2$

Nernst Einstein $\left(u_j = \frac{D_j}{RT} \right)$

$$N_+ = -\nu_+ \left(1 - \frac{z_+}{z_-} \right) D_{H+} \frac{dc}{dz}$$

$-2 \cdot \frac{3}{2}$

$$\boxed{N_{H+} = -3 D_{H+} \frac{dc}{dy}}$$

For binary electrolytes: $t_+ = \frac{z_+ u_+}{z_+ u_+ - z_- u_-} = \frac{z_+ D_+}{z_+ D_+ - z_- D_-}$; $D = \frac{z_+ u_+ D_- - z_- u_- D_+}{z_+ u_+ - z_- u_-}$ or $D = \frac{(z_+ u_+ D_- - z_- u_- D_+) t_-}{(-z_- u_-)}$

$t_- = 1 - t_+$

$= \frac{z_+ u_+ - z_- u_- - z_+ u_+}{z_+ u_+ - z_- u_-}$

$t_- = \frac{(-z_- u_-)}{(z_+ u_+ - z_- u_-)}$

$D = \frac{(z_+ u_+ D_- - z_- u_- D_+) \cdot t_+}{(z_+ u_+)}$

$D = \left(D_- - \frac{z_-}{z_+} D_- \right) t_+$

$D = D_- \left(1 - \frac{z_-}{z_+} \right) t_+$

$= D_- \left(1 - \frac{(-2)}{1} \right) t_+$

$D = 3 D_- t_+$

$D = \left(\frac{-z_+ D_+}{z_- D_-} + D_+ \right) t_-$

$D = D_+ \left(-\frac{z_+}{z_-} + 1 \right) t_- = D_+ \left(\frac{-1}{-2} + 1 \right) t_- = D_+ \left(\frac{3}{2} \right) t_-$

$\therefore \boxed{D = 3 D_- t_+ = \frac{3}{2} D_+ t_-}$

$D_+ = \frac{2D}{3 t_-}$

$$N_{H+} = \frac{-2 \cdot 2}{3 t_-} \frac{\partial c}{\partial y}$$

$$= -\frac{2}{t_-} \frac{\partial c}{\partial y} \quad ; \quad t_- = 1 - t_+$$

$i_y = F \sum z_j \bar{N}_j$

$\frac{i_y}{F} = \sum z_j \bar{N}_j = +1 N_{H+} - 2 \cancel{N_{SO_4^{2-}}}^{0}$ @ $y = 0$

$$\boxed{\frac{i_y}{F} = \frac{-2D}{(1 - t_+)} \frac{\partial c}{\partial y}} \quad @ \; y = 0$$